By Team Orange
Old Dominion
University
CS 410
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Patient Path

“From Fax to Fast: Modernizing Medical Record Exchange”

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Who we are

Kaitlyn Molaison
Project Manager

Kaitlyn is the project manager of PatientPath, contributing to design, development, and documentation while helping guide the team's overall vision and progress.

James Woodson
Software Developer

James is a PatienPath team member who contributes to software design implementation as well as other backend tasks.

Darnell Barnes
Full Stack Developer

Darnell is a PatientPath team member who contributes to the frontend user interface and backend system functionality.

Who we are

Grace Wright
Front End Developer

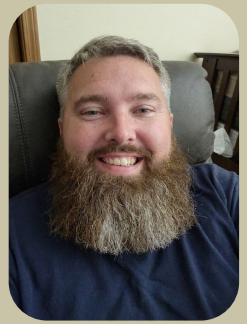
Grace is a PatientPath team member responsible for the front end development. She will contribute to the front end design of the website, and ensure that it is working properly for user's.

Christopher DeHaven
Software Developer

Christopher is a developer for PatientPath who contributes to software design implementation and improving functionality and efficiency within the system.

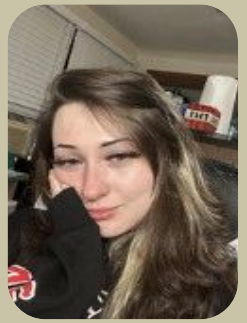
Stephen Garrison
Software Developer

Stephen Garrison is a team member on PatientPath who contributes to system design, technical writing, and building clear project communication through the team website.



Emily Nowak
Documentation Specialist

Emily is a PatientPath team member responsible for documentation and creating process flow diagrams to clearly communicate system design and project goals.



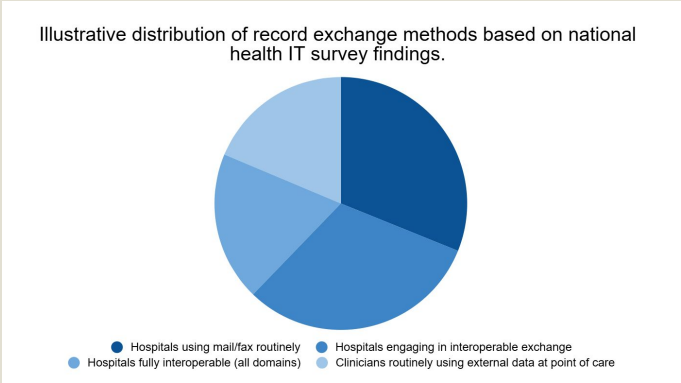
Elevator Pitch

PatientPath is a secure web-based platform that replaces outdated fax systems in healthcare with instant digital record sharing. It allows providers to access and share patient records in real time while integrating with existing EHR (electronic health record) systems. By reducing delays, errors, and administrative work, PatientPath helps deliver faster, safer, and more coordinated care, so patients' medical stories follow them wherever they go.

The Problem

Current medical transfer methods are inefficient and slow.

- 65% of all transfers rely on fax machines despite the availability of digital alternatives
- Slow, unreliable, and prone to errors
  - Issues include lost pages, unreadable documents, and delayed delivery
- Direct impact on patient care
  - Leads to postponed diagnosis and repeated tests
- Disproportionate Harm
  - Especially impactful in rural and underfunded facilities



Office of the National Coordinator for Health Information Technology. (2023). *Interoperable exchange of patient health information among U.S. hospitals: 2023 data brief.* U.S. Department of Health & Human Services. https://www.healthit.gov/data/data-briefs/interoperable-exchange-patient-health-information-among-us-hospitals-2023

Current State of Healthcare IT

Although some healthcare practices have implemented EHR systems, *interoperability still remains a major challenge*.
Key issues include:

- Many EHR systems do not communicate across different organizations.
- Healthcare IT systems remain siloed.
- Smaller and rural communities often lack modern infrastructure.
- Regulatory requirements (HIPAA) increase complexity.

As a result, when systems cannot exchange data directly, providers often default to manual or outdated communication methods.

Source: Office of the National Coordinator for Health Information Technology (2023). Interoperable exchange of patient health information among U.S. hospitals.

Why This Matters

Healthcare is very time sensitive. Delays in information transfer can lead to:

- Postponed specialist appointments
- Repeated diagnostic testing
- Incomplete patient histories
- Increased workload for administrative staff
- All around frustration for patients and providers

In an industry where minutes can affect outcomes, inefficiencies in information exchange have real clinical and financial consequences.

Problem Characteristics

Patient Privacy Risk

Outdated security protocols

Confidential data vulnerability

Disconnected Systems

Limited interoperability

Inefficient Transfer Process

Record delays

Document loss

Illegible fax transmissions

Problem Characteristics

Scalability Barriers

Limited growth adaptability

Rigid architecture

Poor Usability

Cluttered interfaces

Poor navigation

Lack of user roles

Connectivity Constraints

Online-only architecture

Downtime vulnerability

Who is Affected?

- Patients
- Providers
- Admin Staff
- Rural Facilities
- Families/Caregivers

Current Flow Process

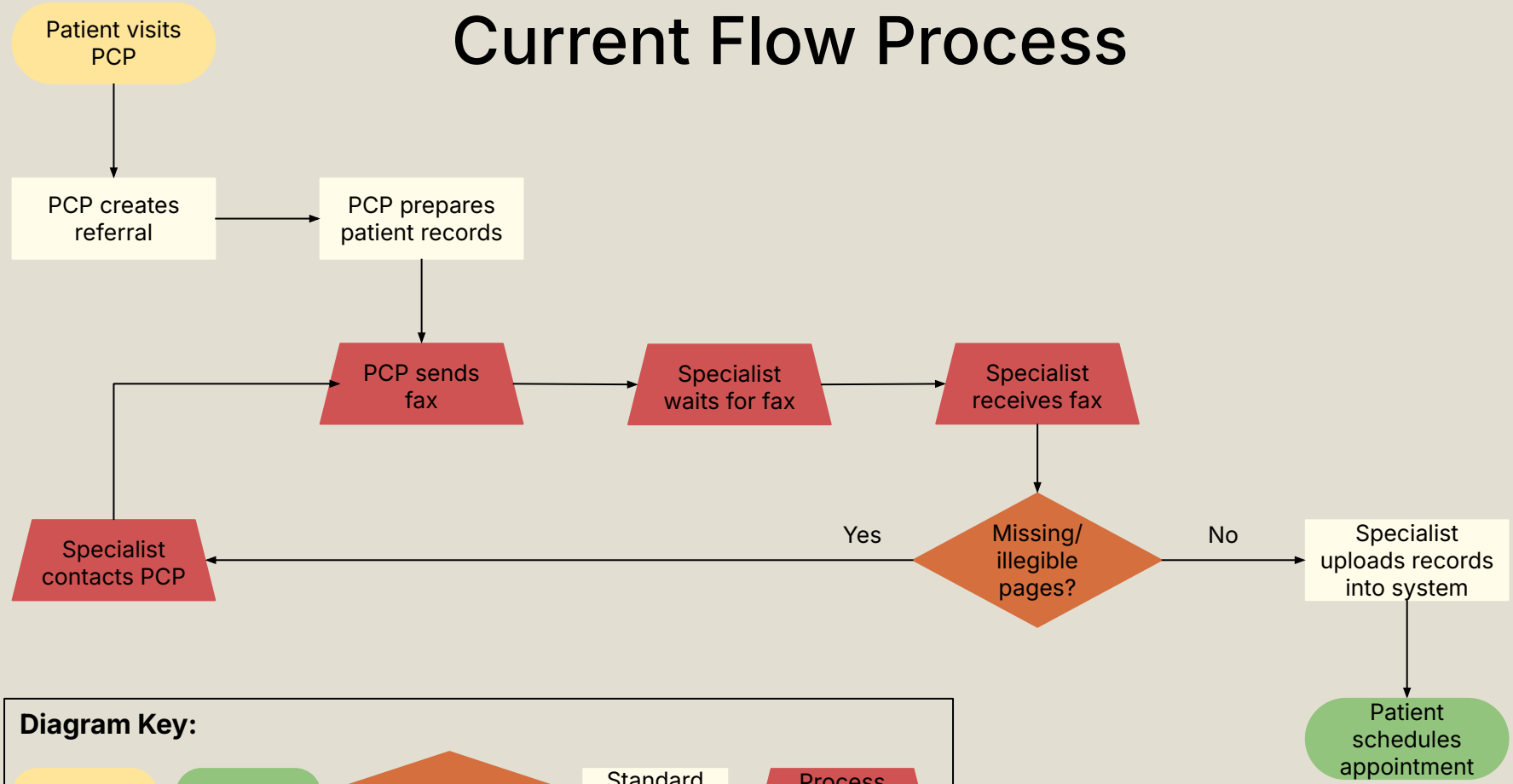


Diagram Key:

Start | Stop | Decision | Standard Step | Process Bottleneck

The Solution

PatientPath is a secure, centralized digital platform designed to replace outdated fax-based communication in healthcare.

Instead of faxing documents, PatientPath enables providers to instantly upload, access, and share medical records through a secure web-based system.

The platform supports hospitals, clinics, private practices, healthcare networks, and their patients.

Unlike traditional EHR systems, PatientPath focuses on secure interoperability and cross-organization communication. It acts as a **secure bridge between systems**, reducing reliance on fax while improving cybersecurity, compliance, and usability.

Solution Characteristics

Security & Compliance

- End-to-end encryption (data in transit + at rest)
- HIPAA-aligned architecture
- Role-Based Access Control (organization-level permissions)
- Multi-factor authentication
- Full audit trail logging

Solution Characteristics

Interoperability & Architecture

- API-first REST architecture
- FHIR & HL7 compatibility
- EHR-agnostic integration (Epic, Oracle, etc.)
- Plugin-based deployment model

Solution Characteristics

Performance & Scalability

- Cloud-hosted (AWS/Azure/GCP compatible)
- 99.9%+ uptime target
- Horizontal scaling architecture
- Offline sync resilience

Solution Process Flow

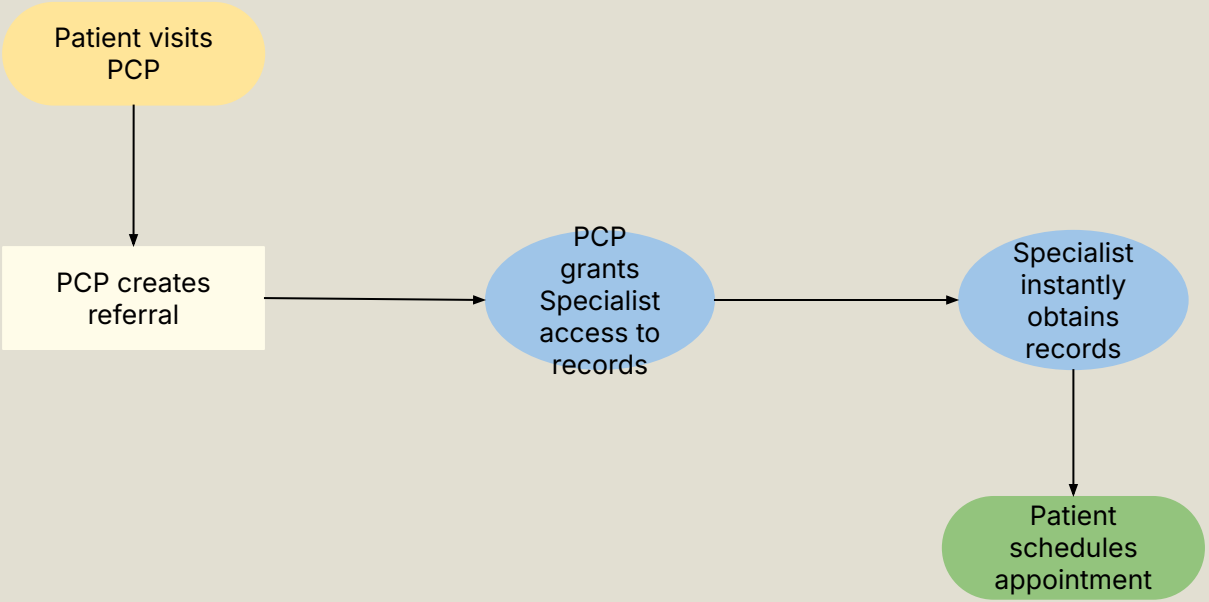


Diagram Key:

Problem–Solution Mapping

	Identified Problem	System Feature	Expected Impact
1	Reliance on fax-based record transfers	Secure real-time digital document exchange	Reduced delays & elimination of document loss
2	Limited interoperability between EHR systems	API-first architecture with FHIR/HL7 compatibility	Standardized cross-organization data exchange
3	Manual referral tracking & follow-up processes	Automated referral workflow with status monitoring	Improved efficiency & faster referral completion
4	Communication methods withsecurity vulnerabilities	End-to-end encryption with RBAC and MFA	Enhanced HIPAA-aligned compliance & access control
5	Connectivity instability in rural / low-resource areas	Offline caching with automatic synchronization	Sustained usability in low-bandwidth environments

Major Functional Components Architecture

Presentation Layer

- Provider Dashboard
- Patient Portal
- Admin Console
- Login & MFA Screens
- Notification Interface

USERS

Patients

Providers

Administrators

Application Layer

- *API Gateway / Web Server*
- Identity & Access Management
- Patient Record Management
- Secure Record Exchange Service
- FHIR / HL7 Integration Service

Data Layer

- PostgreSQL Database (user & organization)
- Encrypted Document Storage (Medical Records)
- Audit Logs
- Consent Records
- FHIR Normalized Data

Patient Path Will

What Patient Path Will Do

- Enable secure digital transfer of medical records between providers
- Provide a centralized hub for accessing and managing patient documents
- Offer real-time tracking of record requests and transfer status
- Support digital patient consent and authorization workflows
- Reduce errors with automated validation and completeness checks
- Maintain HIPAA-compliant audit trails and access controls

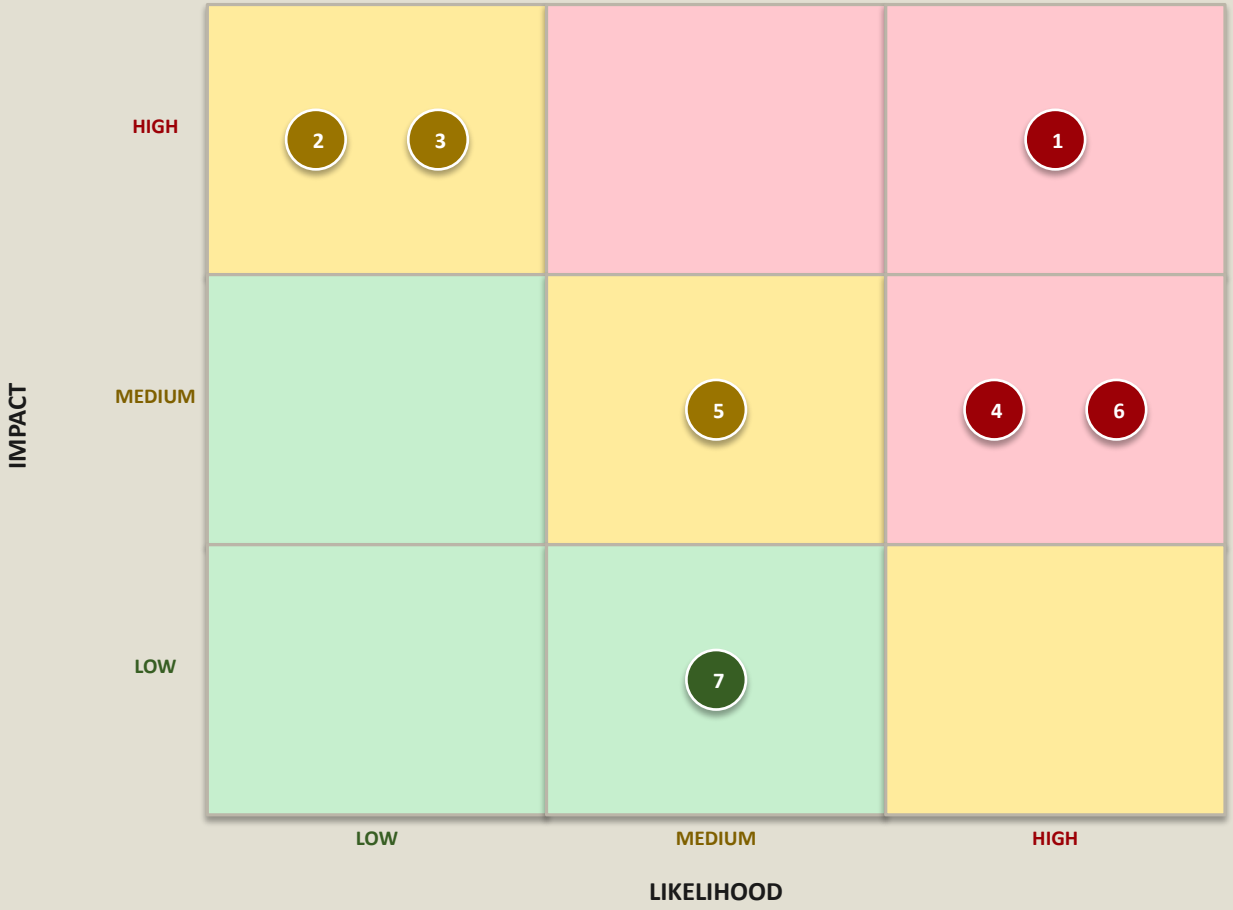
Patient Path Will Not

Provide medical diagnoses or clinical recommendations - The platform does not assist in medical decision making. It’s role is limited to securely transferring and accessing information.

Handle hospital management functions - Scheduling, billing, and internal hospital operations are already handled by specialized systems.

Function as an emergency communication system - The platform supports data exchange, not urgent/emergency situations.

Risk Assessment Matrix



Risk Legend

HIGH	MED	LOW
Technical	Customer	Legal/Sec

1. EHR Integration Complexity
   Technical • Control
2. Cybersecurity / Data Breach
   Legal & Sec • Control
3. HIPAA Non-Compliance
   Legal & Sec • Avoid
4. Provider Adoption Resistance
   Customer • Control
5. System Downtime / Cloud Failure
   Technical • Transfer
6. Scope Creep
   Technical • Avoid
7. Patient Trust / Privacy Concerns
   Customer • Monitor

Risk Mitigation Strategies

Organized by risk category | Approach types: Control · Avoid · Transfer · Monitor

Technical Risks

1 EHR Integration Complexity

Control

Phased FHIR/HL7 rollout with sandbox testing before production deployment

5 System Downtime / Cloud Failure

Transfer

SLA with cloud provider (AWS/Azure) requiring 99.9%+ uptime; failover redundancy

6 Scope Creep

Avoid

Strict stage-gate process with weekly milestone reviews and frozen requirements

Customer Risks

4 Provider Adoption Resistance

Control

Structured onboarding program, intuitive UI design, and workflow alignment workshops

7 Patient Trust / Privacy Concerns

Monitor

Transparent consent flows, clear privacy policy, and regular patient communication

Legal & Security Risks

2 Cybersecurity / Data Breach

Control

End-to-end encryption, MFA, annual penetration testing & incident response plan

3 HIPAA Non-Compliance

Avoid

Legal review at each dev phase; automated compliance checks built into CI/CD pipeline

Market Assessment

Target Segments

- Hospitals and Healthcare networks
- Independent private practices
- Specialty Clinics

Competitive Landscape

- Traditional fax transfers
- Electronic Health Record systems (Epic, Oracle Healthcare, and eClinicWorks)

Competition Comparison Matrix

Capability	PatientPath	Traditional Fax	Kno2 (Digital Fax)	Redox (Healthcare APIs)	CommonWell HIE
Real-time record transfer	✓	✗	Limited	✓	✓
Cross-organization data exchange	✓	✓ (manual)	✓	✓	✓
EHR interoperability (FHIR/HL7)	✓	✗	Limited	✓	✓
Automated referral workflow	✓	✗	Limited	✗	✗
Secure audit logging	✓	✗	✓	✓	✓
Offline support for rural clinics	✓	✓	✗	✗	✗
Ease of deployment	✓	✓	✓	Limited	Limited

Conclusion

PatientPath modernizes healthcare communication by replacing outdated fax-based processes with a secure, efficient digital platform. By improving access to medical records, reducing administrative burden, and strengthening data security, PatientPath helps providers deliver
faster, safer, and more coordinated care. In a healthcare system where time and accuracy matter, PatientPath provides a necessary step forward.

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Glossary

PCP - Primary Care Provider
EHR - Electronic Health Records
HIPAA - Health Insurance Portability and Accountability Act
FHIR - Fast Healthcare Interoperability Resources
RBAC - Role-Based Access Control
HL7 - Health Level Seven